

Network Scanning

Tool: Nmap

Task: Scan a local network device (e.g., Metasploitable2) using `nmap -sV 192.168.1.x`.

Enhanced Tasks

Service Enumeration: Run `nmap -sC -sV 192.168.1.x` to identify services and scripts. Document findings in a table.

Scan Analysis: Compare results of a stealth scan (`-sS`) vs. aggressive scan (`-A`). Summarize differences in a 50-word report.

Proof

Scan Summary 192.168.229.129

scan local network machine using

CMD: `nmap -sV 192.168.223.129`

```
(root@kali)-[/home/blackhat/task]
# nmap -sV 192.168.229.129
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-06 11:29 +0530
Nmap scan report for 192.168.229.129
Host is up (0.0018s latency).
```

Enumeration

CMD: `nmap -sC -sV 192.168.223.129`

```
(root@kali)-[/home/blackhat/task]
# nmap -sC -sV 192.168.229.129
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-06 11:44 +0530
Nmap scan report for 192.168.229.129
Host is up (0.0021s latency).
```

Findings

PORT	STATE	SERVICE	VERSION
21	Open	FTP	vsftpd 2.3.4
22	Open	SSH	OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23	Open	TELNET	Linux telnetd
25	Open	SMTP	Postfix smtpd
53	Open	DOMAIN	ISC BIND 9.4.2
80	Open	HTTP	Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111	Open	RPCBIND	2 (RPC #100000)
139	Open	NETBIOS-SSN	Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445	Open	NETBIOS-SSN	Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512	Open	EXEC	netkit-rsh rexecd
513	Open	LOGIN	OpenBSD or Solaris rlogind
1099	Open	JAVA-RMI	GNU Classpath grmiregistry
1524	Open	BINDSHELL	Metasploitable root shell
2049	Open	NFSS	2-4 (RPC #100003)
3306	Open	MYSQL	ProFTPD 1.3.1
2121	Open	FTP	MySQL 5.0.51a-3ubuntu5
5432	Open	POSTGRESQL	PostgreSQL DB 8.3.0 - 8.3.7
5900	Open	VNC	VNC (protocol 3.3)
6000	Open	x11	(access denied)
6667	Open	IRC	UnrealIRCd
8009	Open	AJP13	Apache Jserv (Protocol v1.3)
8189	Open	HTTP	Apache Tomcat/Coyote JSP engine 1.1

Scan Analysis

Stealth Scan (-sS) is a fast, half-open port-scanning method that determines if a port is open without completing the full TCP three-way handshake.

```
(root@kali)-[/home/blackhat/task]
# nmap -sS 192.168.229.129
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-06 14:48 +0530
Nmap scan report for 192.168.229.129
Host is up (0.0027s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:38:11:69 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.81 seconds
```

Aggressive Scan (-A) is a shortcut command that combines multiple advanced reconnaissance options into a single action to gather deep information quickly.

```
(root@kali)-[/home/blackhat/task]
# nmap -A 192.168.229.129
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-06 14:50 +0530
Nmap scan report for 192.168.229.129
Host is up (0.00038s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|   STAT:
|   FTP server status:
|     Connected to 192.168.229.128
|     Logged in as ftp
|     TYPE: ASCII
|     No session bandwidth limit
|     Session timeout in seconds is 300
|     Control connection is plain text
|     Data connections will be plain text
|     vsFTPD 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
```

```

25/tcp open smtp Postfix smtpd
|_ssl-date: 2026-09-06T09:20:53+00:00; +8s from scanner time.
|_sslv2:
|   SSLv2 supported
|   ciphers:
|     SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|     SSL2_DES_192_EDE3_CBC_WITH_MD5
|     SSL2_RC4_128_EXPORT40_WITH_MD5
|     SSL2_RC2_128_CBC_WITH_MD5
|     SSL2_RC4_128_WITH_MD5
|     SSL2_DES_64_CBC_WITH_MD5
|_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
|_smtp-command: metasploitable.localdomain, PIPELINING, SIZE 1024000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
53/tcp open domain ISC BIND 9.4.2
|_dns-nsid:
|_bind.version: 9.4.2
80/tcp open http Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_http-title: Metasploitable2 - Linux
111/tcp open rpcbind 2 (RPC #100000)
|_rpcinfo:
|   program version port/proto service
|   100000 2 111/tcp rpcbind
|   100000 2 111/udp rpcbind
|   100003 2,3,4 2049/tcp nfs
|   100003 2,3,4 2049/udp nfs
|   100005 1,2,3 50426/tcp mountd
|   100005 1,2,3 53046/udp mountd
|   100021 1,3,4 32964/tcp nlockmgr
|   100021 1,3,4 52558/udp nlockmgr
|   100024 1 35516/udp status
|   100024 1 50978/tcp status
139/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp open netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp open exec netkit-rsh rshcd
513/tcp open login?

```

```

514/tcp open tcpwrapped
1099/tcp open java-rmi GNU Classpath grmiregistry
1524/tcp open bindshell Metasploitable root shell
2049/tcp open nfs 2-4 (RPC #100003)
2121/tcp open ftp ProFTPD 1.3.1
3306/tcp open mysql MySQL 5.0.51a-3ubuntu5
|_mysql-info:
|   Protocol: 10
|   Version: 5.0.51a-3ubuntu5
|   Thread ID: 39
|   Capabilities flags: 43564
|   Some Capabilities: Speaks41ProtocolNew, LongColumnFlag, Support41Auth, SwitchToSSLAfterHandshake, SupportsTransactions, SupportsCompression, ConnectWithDatabase
|   Status: Autocommit
|   Salt: }?>[.s{0y5;fq(nrq`n
5432/tcp open postgresql PostgreSQL DB 8.3.0 - 8.3.7
|_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
|_ssl-date: 2026-09-06T09:20:53+00:00; +8s from scanner time.
5900/tcp open vnc VNC (protocol 3.3)
|_vnc-info:
|   Protocol version: 3.3
|   Security types:
|     VNC Authentication (2)
6000/tcp open X11 (access denied)
6667/tcp open irc UnrealIRCd
|_irc-info:
|   users: 1
|   servers: 1
|   lusers: 1
|   lservers: 0
|   server: irc.Metasploitable.LAN
|   version: Unreal3.2.8.1. irc.Metasploitable.LAN
|   uptime: 0 days, 3:29:48
|   source ident: nmap
|   source host: 8FAA09BA.2E4F395.FFFA6D49.IP
|_error: Closing Link: uyweyfgbo[192.168.229.128] (Quit: uyweyfgbo)

```

```
8009/tcp open  ajp13      Apache Jserv (Protocol v1.3)
|_ajp-methods: Failed to get a valid response for the OPTION request
8180/tcp open  http       Apache Tomcat/Coyote JSP engine 1.1
|_http-favicon: Apache Tomcat
|_http-server-header: Apache-Coyote/1.1
|_http-title: Apache Tomcat/5.5
MAC Address: 00:0C:29:38:11:69 (VMware)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
|_ smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_ message_signing: disabled (dangerous, but default)
|_ smb2-time: Protocol negotiation failed (SMB2)
|_ clock-skew: mean: 1h00m08s, deviation: 2h00m00s, median: 7s
|_ nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
|_ smb-os-discovery:
|   OS: Unix (Samba 3.0.20-Debian)
|   Computer name: metasploitable
|   NetBIOS computer name:
|   Domain name: localdomain
|   FQDN: metasploitable.localdomain
|_ System time: 2026-09-06T05:20:44-04:00

TRACEROUTE
HOP RTT      ADDRESS
1   0.38 ms  192.168.229.129

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 27.29 seconds
```

Summary

The stealth scan (-sS) identified open ports and basic services quickly, completing in 0.81 seconds. The aggressive scan (-A) provided significantly more information, including service versions, operating system details, SMB information, SSL details, scripts, hostnames, and traceroute. However, it required 27.29 seconds, making it slower and noisier.